

Life Expectancy

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Problem Statment

To what extent that the socioeconomic factors could affect individual life expectancy in the US?

Dataset Description and Document

The dataset is obtained from this github repo: <https://github.com/mi3nts/life-expectancy>. The author collected and compiled life expectancy and its corresponding demographic data according to different counties, from 2012 to 2019.

The data is collected and combined from the three sources below:

- (1) Demographic and socioeconomic data from US Census and related sources
- (2) Atmospheric reanalysis data from CAMS (Copernicus Atmosphere Monitoring Service)
- (3) Environmental variables from ECMWF ERA5 reanalysis

The original dataset contains 15 socioeconomic features and 87 environmental variables. However, since our question only focus on the impact of socioeconomic factors, we will just use the 15 socioeconomic features for our analysis.

The dataset contains data from 2012 to 2019. For our preliminary analysis, we will only focus on **2019** data.

Full description about the data columns are available in the same github repo: <https://github.com/mi3nts/life-expectancy>.

Loading Dataset

```
# Full dataset with socioeconomic + environmental features
full.data <- read.csv("final_combined_all_variables.csv")
dim(full.data)
[1] 24487 131
```

```
# Filter the data Extract only the first 19 socioeconomic variables and 2019
# data
row.extract <- which(full.data$Year == 2019)
life.data <- full.data[row.extract, 1:19]
```

```
# Clean up the long column name
colnames(life.data) <- c(
  "county", "state", "fips",           # Identifiers
  "life_exp",                          # Target
  "poverty_rate", "edu_hs", "edu_bach", # Socioeconomic
  "median_age", "gini_index", "disability",
```

```

"total_pop", "median_income", "unemployment",
"pop_white", "pop_hispanic", "pop_black",
"no_vehicle", "rent_burden", "single_mother" # Housing/Transport
)

# Quick check to ensure no weird symbols remain
colnames(life.data) <- make.names(colnames(life.data))

# Preview your clean names
head(life.data)

```

	county	state	fips	life_exp	poverty_rate	edu_hs	edu_bach
21427	Autauga County	Alabama	76.59327	15.2	88.5	26.6	38.2
21428	Baldwin County	Alabama	78.29352	10.4	90.8	31.9	43.0
21429	Barbour County	Alabama	74.81453	30.7	73.2	11.6	40.4
21430	Bibb County	Alabama	73.92005	18.1	79.1	10.4	40.9
21431	Blount County	Alabama	75.02919	13.6	80.5	13.1	40.7
21432	Bullock County	Alabama	73.94933	28.9	74.7	12.1	40.2

	median_age	gini_index	disability	total_pop	median_income	unemployment
21427	0.4542	19.0	55380	58731	3.7	2019
21428	0.4587	14.2	212830	58320	4.3	2019
21429	0.4883	21.4	25361	32525	9.2	2019
21430	0.4487	17.2	22493	47542	7.3	2019
21431	0.4570	14.1	57681	49358	3.4	2019
21432	0.4731	12.0	10248	37785	3.9	2019

	pop_white	pop_hispanic	pop_black	no_vehicle	rent_burden	single_mother
21427	76.79126	2.825930	19.028530	4.397813	25.24934	8.052534
21428	86.20542	4.562797	9.264671	3.281849	18.92782	6.748088
21429	46.80021	4.357084	47.576988	10.165864	20.66229	15.403265
21430	76.78833	2.574134	22.291380	6.167465	21.49745	6.285237
21431	95.45951	9.261282	1.608849	4.125294	15.43852	6.252521
21432	21.89696	2.556596	74.804840	10.707185	28.71190	16.253994

```

dim(life.data)
[1] 3061 19

# Inspect the structure of the data
str(life.data)
'data.frame': 3061 obs. of 19 variables:
 $ county      : chr  "Autauga County" "Baldwin County" "Barbour County" "Bibb County" ...
 $ state       : chr  "Alabama" "Alabama" "Alabama" "Alabama" ...
 $ fips        : num  76.6 78.3 74.8 73.9 75 ...
 $ life_exp    : num  15.2 10.4 30.7 18.1 13.6 28.9 22.5 17.9 17.3 13.9 ...
 $ poverty_rate : num  88.5 90.8 73.2 79.1 80.5 74.7 85 84.4 81.6 81.6 ...
 $ edu_hs      : num  26.6 31.9 11.6 10.4 13.1 12.1 16.1 18.5 13.3 12.8 ...
 $ edu_bach    : num  38.2 43 40.4 40.9 40.7 40.2 40.8 39.6 42 46.5 ...
 $ median_age  : num  0.454 0.459 0.488 0.449 0.457 ...
 $ gini_index  : num  19 14.2 21.4 17.2 14.1 12 16.5 21.2 18.1 17.8 ...
 $ disability  : num  55380 212830 25361 22493 57681 ...
 $ total_pop   : num  58731 58320 32525 47542 49358 ...
 $ median_income: num  3.7 4.3 9.2 7.3 3.4 3.9 6.8 8.1 3.9 4.4 ...
 $ unemployment: int  2019 2019 2019 2019 2019 2019 2019 2019 2019 2019 ...
 $ pop_white   : num  76.8 86.2 46.8 76.8 95.5 ...
 $ pop_hispanic: num  2.83 4.56 4.36 2.57 9.26 ...
 $ pop_black   : num  19.03 9.26 47.58 22.29 1.61 ...
 $ no_vehicle  : num  4.4 3.28 10.17 6.17 4.13 ...

```

```
$ rent_burden : num  25.2 18.9 20.7 21.5 15.4 ...
$ single_mother: num   8.05 6.75 15.4 6.29 6.25 ...

# Export the dataset
write.csv(life.data, "2019_life_socio.csv", row.names = FALSE)
```